



Academy of
Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

School of E&TC Engineering

TY-BTech Major Project (SEM-V)

Project –I(2307391)

Automated diagnosis of schizophrenia using ML techniques

Vishvas Raina 202301060005

Piyush Raundal 202301060043

Nachiket Mahajan 202301060047

Sahil Raichurkar 202301060051

Group No :- 66

**Guided By :-
Dr. Vipin Yadav**

Index

- Problem Statement and Objective
- What is Schizophrenia?
- Why do we need to automate the Diagnosis of Schizophrenia?
- Literature Survey
- Difference in EEG Signals between a Normal Person and a Schizophrenic Person
- Block Diagram
- EEG Database
- Method A
- Software Method of Bipolarisation
- Features of Double Banana
- Montage and Technicalities
- Flow Diagram
- Feature Extraction
- Feature Selection
- Feature Selection of Whole Brain
- Regions of Brain

Index

- Region Wise Feature Selection
- Model Selection and Metrics
- Results: Whole Brain Analysis
- Results: Region Wise Analysis - Correlation
- Results: Region Wise Analysis - PCA
- Summary of Results
- Method B
- Variational Mode Decomposition
- Wrapper Based Feature Selection
- Optimistaion techniques(FOA & ABC)
- Results: Whole Brain Analysis(FOA & ABC)
- Results: Region Wise Analysis - FOA
- Results: Region Wise Analysis - ABC
- Summary of Results
- SDG Goals
- References

Problem Statement

To develop and evaluate a machine learning model for the classification of schizophrenia based on features extracted from pre-processed, multi-channel EEG signals.

Objectives

- To preprocess EEG signals (artifact removal, filtering, normalization).
- To identify the effective features to capture the schizophrenia pattern in EEG signal
- To design a EEG based Schizophrenia diagnosis framework based on ML.

Scope of the Project

- This method could help psychiatrists detect schizophrenia earlier and more objectively.
- It can be used in hospital tools allowing for constant monitoring.
- Neurologists who interpret EEG scans could use this tool to screen for neurophysiological markers associated with schizophrenia

What is Schizophrenia?

- Schizophrenia is a chronic brain disorder that affects how a person thinks, feels, and perceives reality.
- Characterized by distorted thoughts, hallucinations (hearing/seeing things), delusions (false beliefs), and disorganized behavior.
- It is not split personality, but a disorder of disrupted brain communication and perception.



Fig.1. Symptoms of Schizophrenia

Why do we need to automate the diagnosis of Schizophrenia?

- Symptom overlap with depression, bipolar disorder, and drug-induced psychosis leads to frequent misdiagnosis.
- Current diagnosis depends on subjective interviews and observation.
- EEG provides objective biomarkers that can improve reliability.
- Shortage of psychiatrists, especially in rural areas, causes delays and long waiting times.

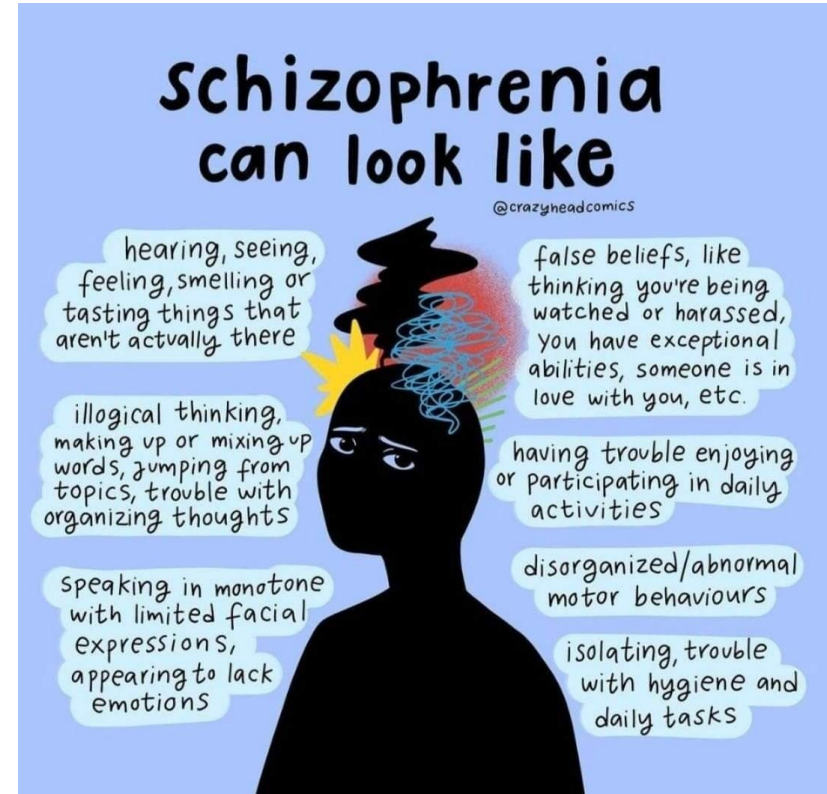


Fig.2. What goes through the brain

Literature Survey

Sl. No.	Authors and Year	Title of Paper/Article	Objective	Methodology/ Approach	Key Findings	Limitations/ Gaps	Future Scope
1	Heda et al. (2024) [7]	miRNA-Based Diagnosis of Schizophrenia Using Machine Learning	To use miRNA biomarkers and ML to create a more reliable diagnostic tool for schizophrenia.	Integrates miRNA and EEG data. Utilizes advanced feature extraction (CNNs) and emphasizes Explainable AI (XAI) for clinical trust.	High accuracy (90-98%) can be achieved with deep learning on EEG data. Combining biological markers (miRNA) with ML is a highly promising approach.	Symptom heterogeneity, small/noisy datasets, and a lack of large, standardized studies limit model generalizability.	Combine multimodal data (miRNA, EEG, MRI), develop interpretable AI models for clinical use, and conduct large-scale longitudinal studies.
2	Vyškovský et al. (2022) [8]	Structural MRI-Based Schizophrenia Classification Using Autoencoders and 3D Convolutional Neural Networks...	To test if deep learning models (SAE and 3D CNNs) can classify schizophrenia using structural MRI data and evaluate different preprocessing techniques.	Used sMRI from 104 subjects with various preprocessing (VBM, DBM). Trained Stacked Autoencoders (SAE) on selected features and 3D CNNs on whole images.	Stacked Autoencoders on VBM-processed data performed best (~70% accuracy). 3D CNNs were less effective, and combining features did not help, likely due to the small dataset.	Very small dataset (N=104) limits the use of complex models like CNNs. Lack of external validation and model interpretability.	Use larger, multi-center datasets. Focus on model interpretability and integrate multimodal data (e.g., MRI + EEG) for better accuracy.
3	Thilagavathi et al. (2024) [9]	Schizophrenia Detection Using Machine Learning Approach	To use a Support Vector Machine (SVM) on extracted EEG features to differentiate schizophrenia patients from healthy controls.	Analyzed EEG data from a small cohort (14 patients, 14 controls). Extracted spectral and entropy features and fed them into an SVM classifier.	The SVM model showed high accuracy, demonstrating that standard ML can effectively classify schizophrenia on small, balanced datasets, serving as a supplementary diagnostic tool.	Extremely small dataset (N=28) severely limits the generalizability of the results. Medication effects were not controlled for.	Validate the approach on larger, more diverse datasets to confirm its effectiveness and test its robustness for real-world application.

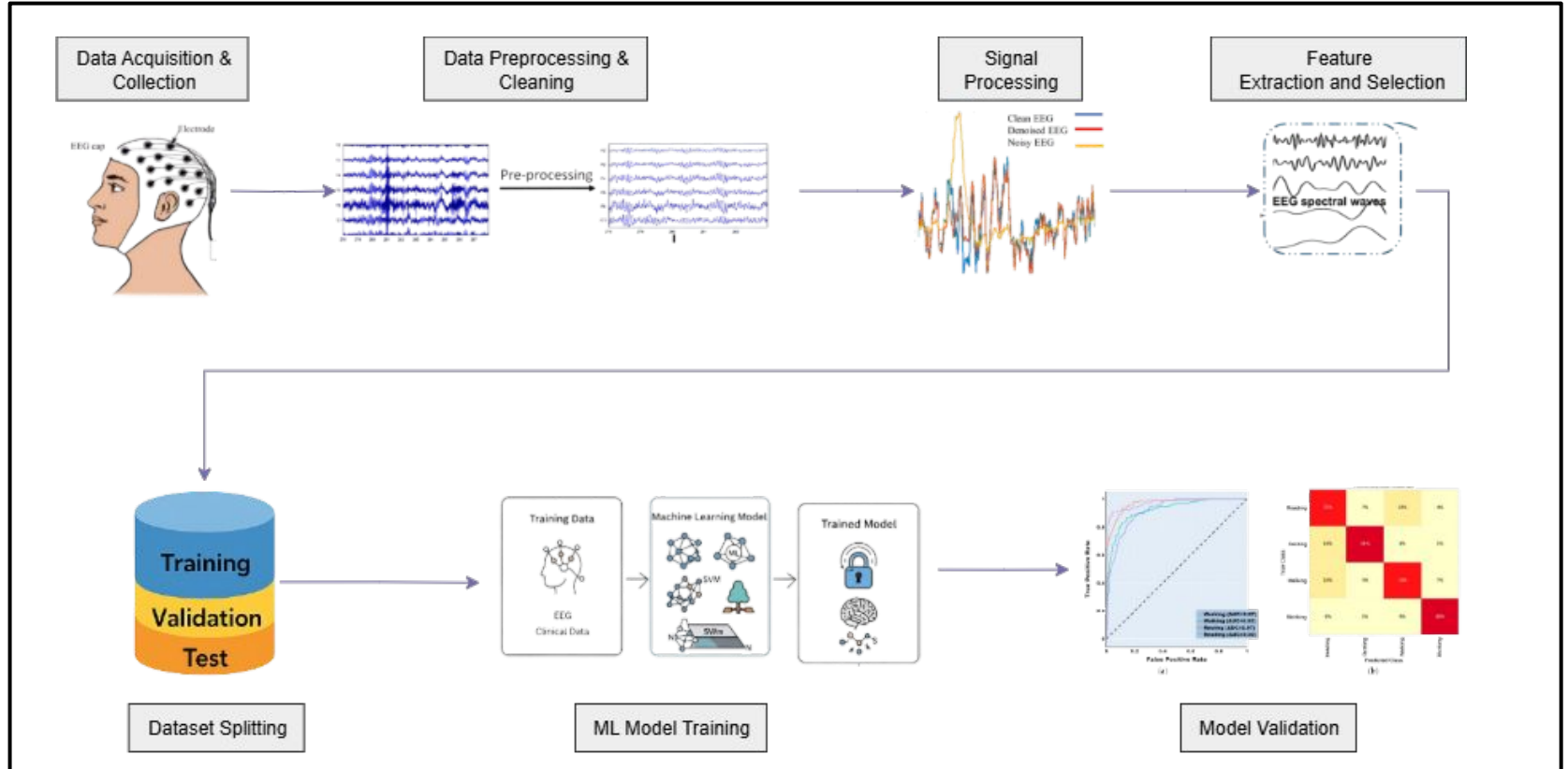
Literature Survey

Sl. No.	Author(s) and Year	Title of Paper/Article	Objective	Methodology/ Approach	Key Findings	Limitations/ Gaps	Future Scope
4	Yamashita et al. (2021) [10]	Three-Dimensional Convolutional Autoencoder Extracts Features of Structural Brain Images With a 'Diagnostic Label-Free' Approach...	To use a 3D convolutional autoencoder (3D-CAE) to extract features from sMRI scans without using diagnostic labels during the learning phase.	Trained a 3D-CAE on two separate MRI datasets (Kyoto and COBRE). Used the extracted features to predict clinical severity and medication dosage via regression.	The label-free 3D-CAE approach successfully extracted features that were highly predictive of clinical variables (symptoms, medication). It outperformed traditional methods.	The model's architecture was not fully optimized. The approach needs testing on other psychiatric disorders and with longitudinal data.	Extend to other disorders, incorporate longitudinal and multimodal data (EEG, genetics), and improve model interpretability for clinical use.
5	Uhlhaas & Singer (2010) [5]	Abnormal Neural Oscillations and Synchrony in Schizophrenia	To review and synthesize evidence linking neural oscillation abnormalities across different frequency bands to the cognitive deficits seen in schizophrenia.	A narrative literature review summarizing findings from human electrophysiology (EEG/MEG) and animal model studies to build a conceptual framework.	Schizophrenia is associated with widespread dysregulation of both high (gamma) and low-frequency oscillations. Disrupted cross-frequency coupling is likely a key mechanism behind cognitive dysfunction.	Findings across the literature are often contradictory and difficult to compare due to varying methodologies. The review can only infer, not establish, causal links.	Systematically study cross-frequency interactions, use mechanistic models to understand the neurobiological basis (e.g., GABA dysfunction), and standardize methods for biomarker development.

Difference in EEG Signals between a normal person and a schizophrenic person

	Normal Person	Schizophrenic Person
Delta(< 4 Hz)	Deep sleep; rare when awake	Elevated during wakefulness; altered sleep delta
Theta(4–8 Hz)	Drowsiness, meditation	Elevated resting/task-related theta
Alpha(8–12 Hz)	Relaxed wakefulness (eyes closed)	Reduced alpha power and coherence, especially in medicated patients; associated with negative symptoms.
Beta(12–30 Hz)	Active thinking, alertness	Mixed often elevated in resting, eyes open
Gamma(>30 Hz)	Cognitive integration, memory	Dysregulated: power and synchrony abnormalities. Reduced gamma power and synchronization during tasks (sensory gating, working memory); sometimes increased at rest.

Block Diagram



EEG Database

- There are two EEG data archives for two groups of subjects. The subjects were adolescents who had been screened by psychiatrist and divided into two groups: healthy ($n = 39$) and with symptoms of schizophrenia ($n = 45$).
- Each file contains EEG data for one subject, with samples from 16 channels listed sequentially, where each value represents an EEG amplitude in microvolts
- First 7680 samples represent 1st channel, then 7680 - 2nd channel, etc. The sampling rate is 128 Hz, thus 7680 samples refer to 1 minute of EEG record.

1 - F7	9 - T4
2 - F3	10 - T5
3 - F4	11 - P3
4 - F8	12 - Pz
5 - T3	13 - P4
6 - C3	14 - T6
7 - Cz	15 - O1
8 - C4	16 - O2

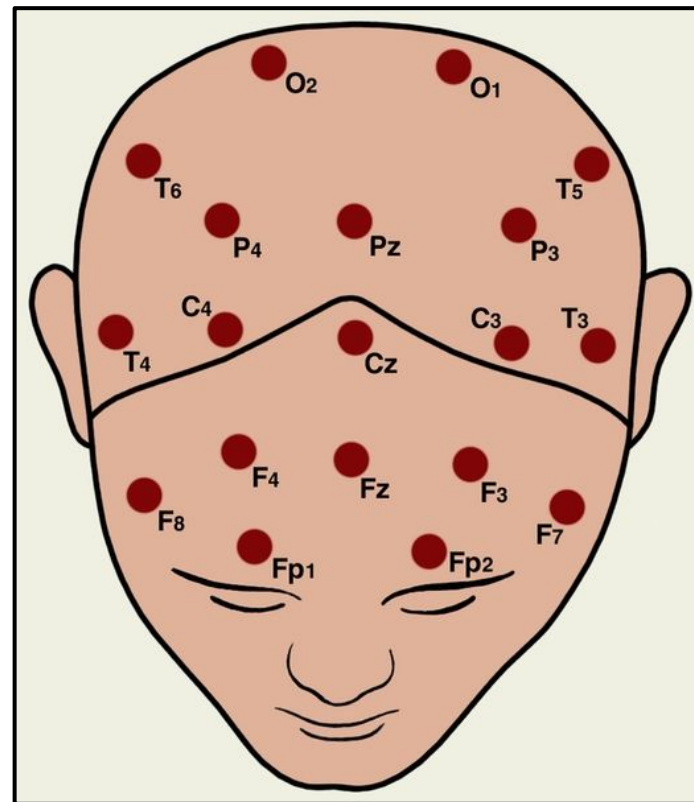
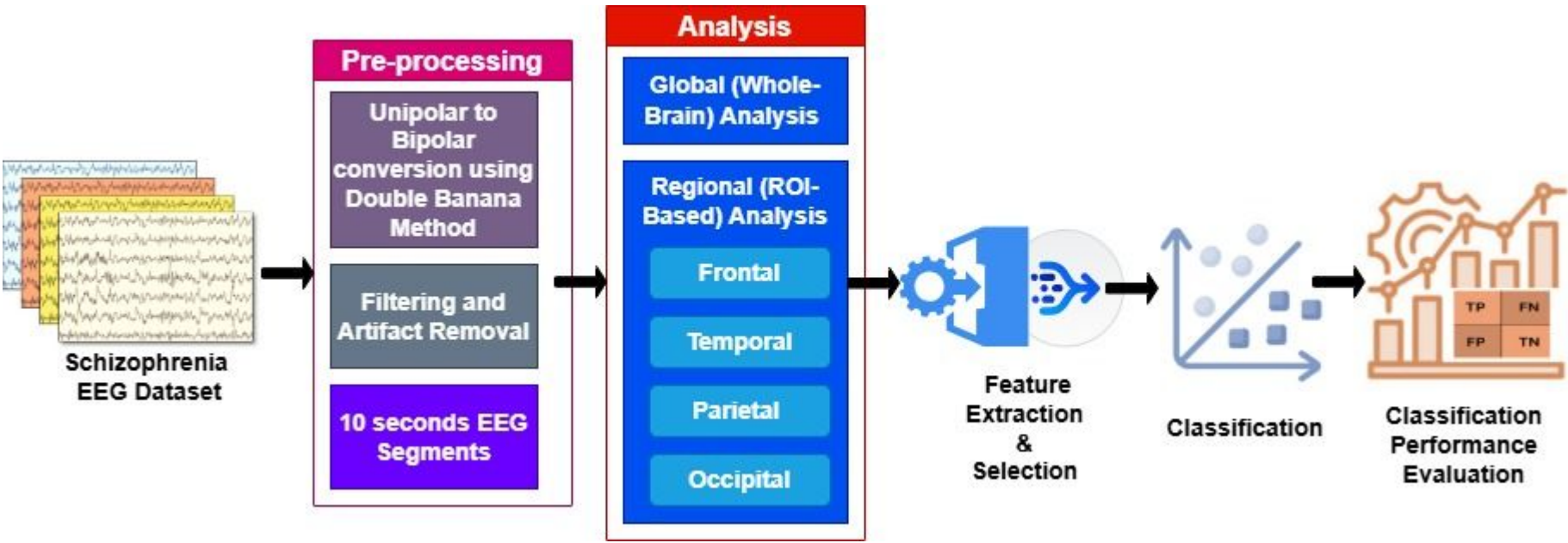


Fig.3. Electrode Placement System

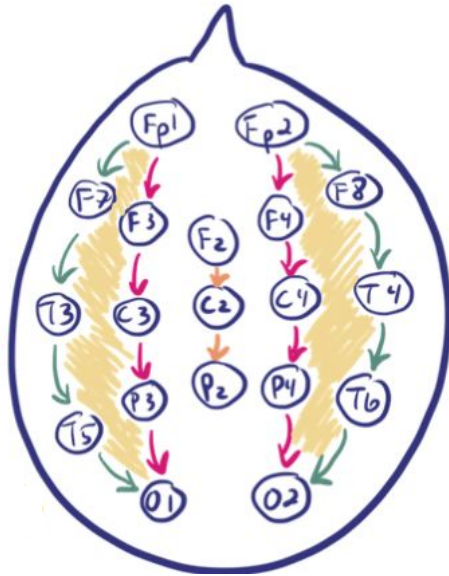
Method A

Flow Diagram

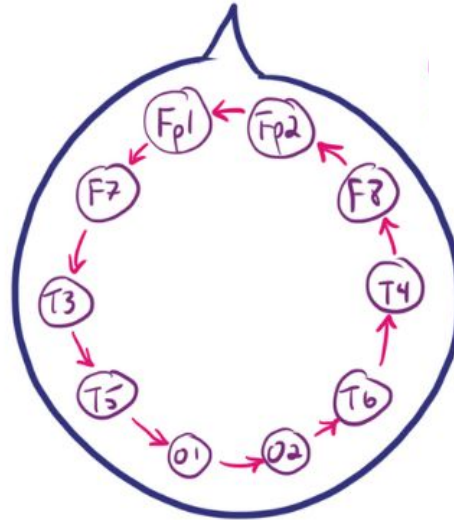


Software Method of Bipolarisation

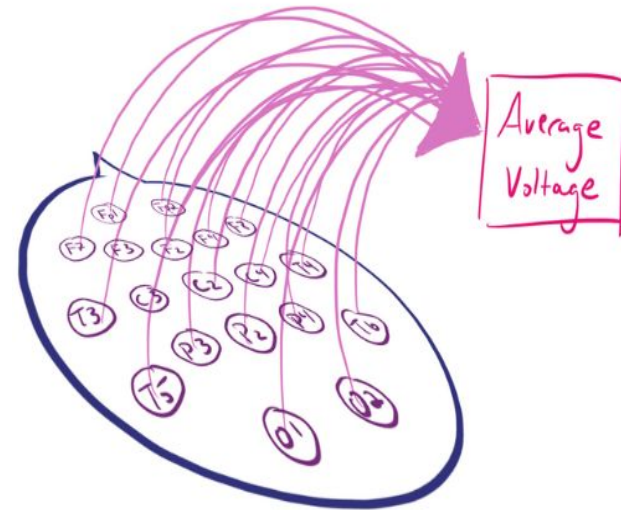
Double Banana



Circumferential

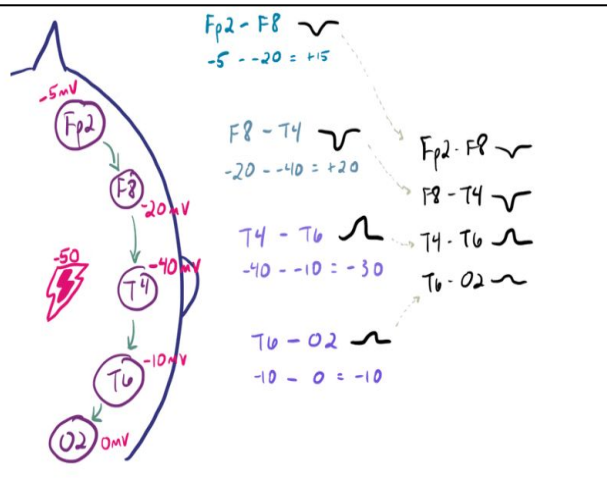


Referential

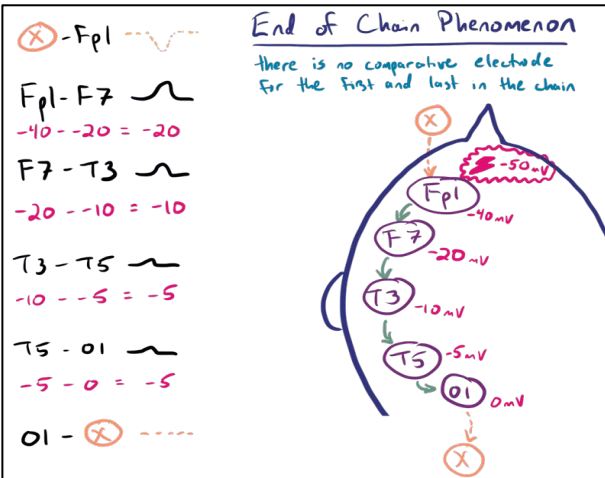


Features of Double Banana

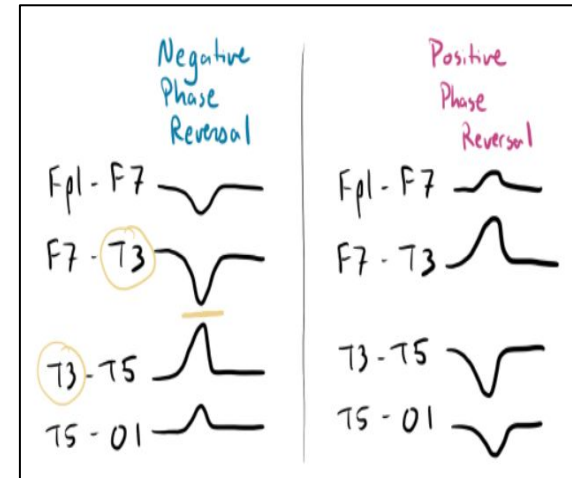
EEG Tracing



End of Chain Phenomenon

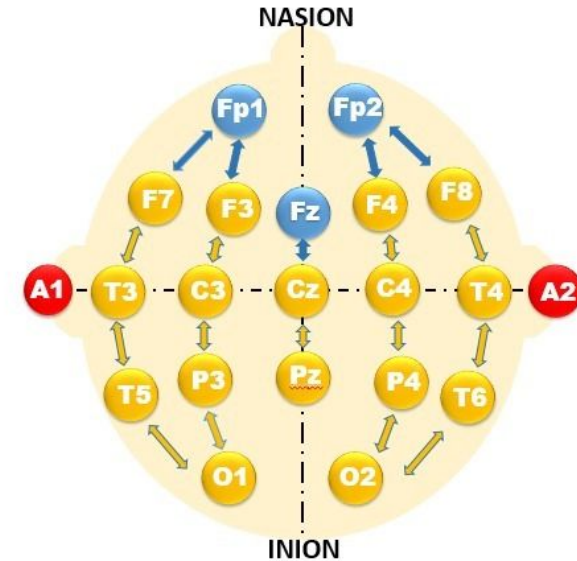


Phase Reversal



Montage and Technicalities

Left hemisphere		Central Pairs	Right hemisphere	
Left Temporal chain	Left Parasagittal chain		Right Temporal chain	Right parasagittal chain
F7-T3	F3-C3	Cz-Pz	F8-T4	F4-C4
T3-T5	C3-P3		T4-T6	C4-P4
T5-O1	P3-O1		T6-O2	P4-O2



Feature Extraction

- EEG data from all 16 channels was divided into 10-second segments, and features were extracted using Jingwei Too's MATLAB EEG Feature Extraction Toolbox.
- Although the toolbox provides 30 methods, many return multiple values, resulting in 429 total features per segment.
- Since 429 features are too many for efficient classification, feature selection was needed to reduce dimensionality and keep only the most relevant features; the same process will be applied to 5-second and 2-second segments.

Mean Curve Length (mcl)	Log Root Sum of Sequential Variation (lrssv)	Minimum (min)
Hjorth Activity (ha)	Tsallis Entropy (te)	Autoregressive Model (ar)
Hjorth Mobility (hm)	Shannon Entropy (sh)	Kurtosis (kurt)
Hjorth Complexity (hc)	Log Energy Entropy (le)	Skewness (skew)
First Difference (1d)	Renyi Entropy (re)	Bandpower Delta (bpd)
Normalized First Difference (n1d)	Arithmetic Mean (am)	Bandpower Theta (bpt)
Second Difference (2d)	Standard Deviation (sd)	Bandpower Alpha (bpa)
Normalized Second Difference (n2d)	Variance (var)	Bandpower Beta (bpb)
Mean Energy (me)	Median (md)	Bandpower Gamma (bpg)
Mean Teager Energy (mte)	Maximum (max)	Ratio Bandpower Alpha/Beta (rba)

Feature Selection

1. Correlation Matrix:

- Computed the correlation matrix for all features.
- Identified highly correlated pairs (> 0.9), indicating redundancy.
- Compared variances within each pair and kept the feature with higher variance.
- Removed the lower-variance feature to reduce redundancy.
- This step preserved the most informative features while simplifying the feature set.

2. PCA Transformation:

- Used PCA to reduce the number of features.
- Converted the original features into new, independent components.
- Sorted these components based on how much variance they captured.
- Kept only the components that together explained 95% of the total variance.

Feature Selection of Whole Brain

- Extracted 429 features from the 10-second EEG segments.
- Computed the correlation matrix to identify highly correlated feature pairs (correlation > 0.9).
- Treated such pairs as redundant since they convey overlapping information.
- For each correlated pair, compared their variances and retained the feature with higher variance, as it likely offers better discriminative power.
- Removed the lower-variance feature from each pair.
- This redundancy-removal step reduced the feature set from 429 features to 254 features.

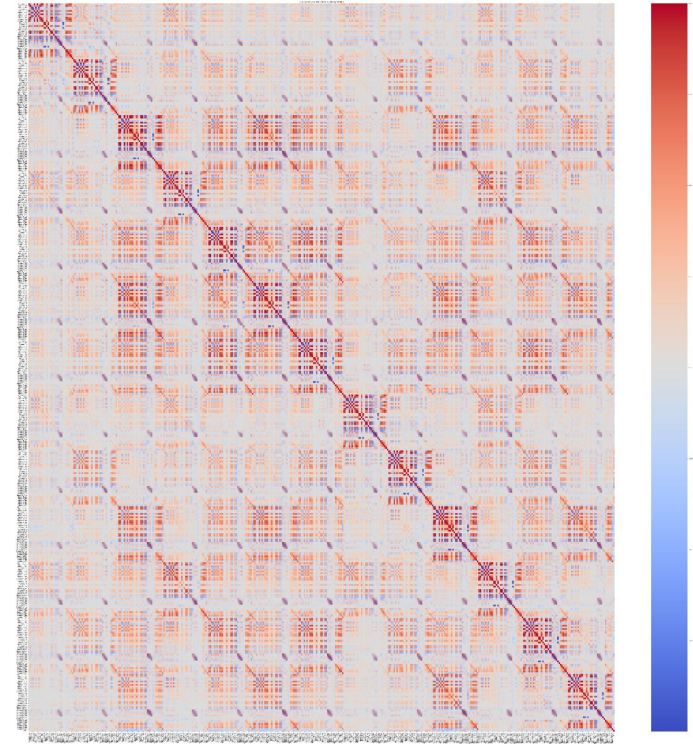
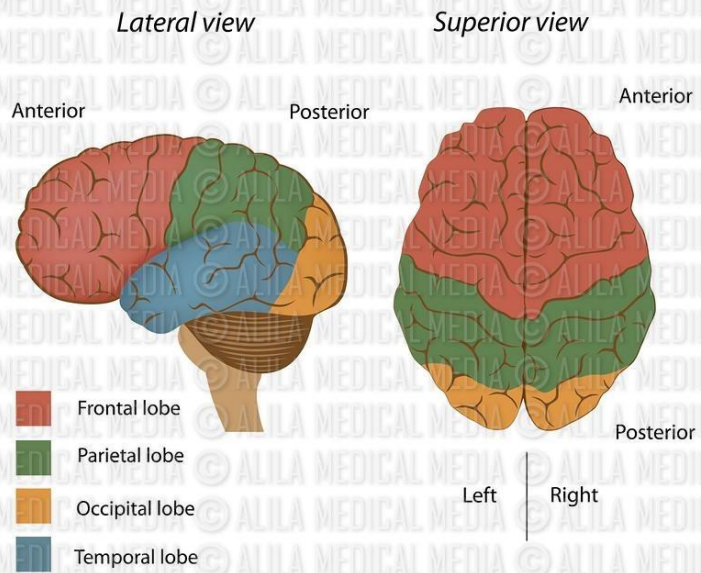


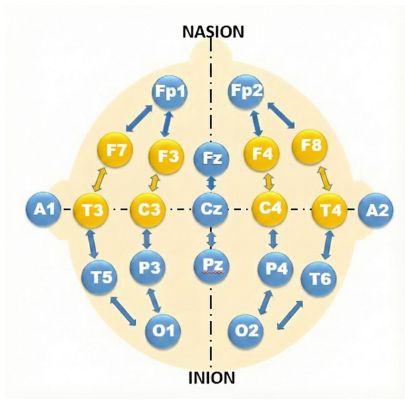
Fig 4. Correlation Matrix

Regions of Brain

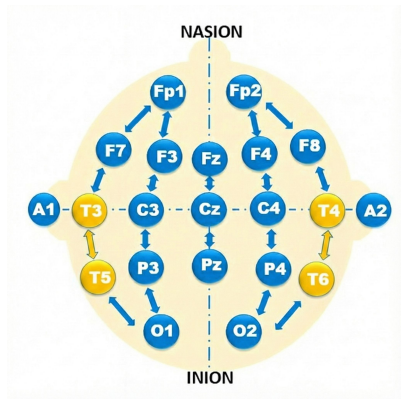
Lobes of the Brain



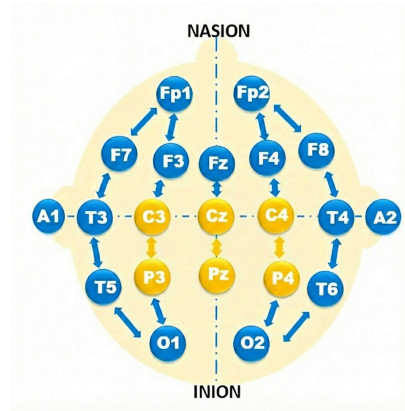
© Alila Medical Media - www.AlilaMedicalMedia.com



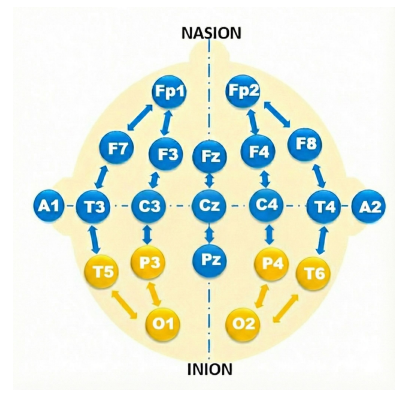
Frontal Region



Temporal Region



Parietal Region



Occipital Region

Fig 5. Regions of the Brain

Region Wise Feature Selection

Frontal Region:

1. Left Frontal:

- a. Bipolar 1: F7-T3
- b. Bipolar 4: F3-C3

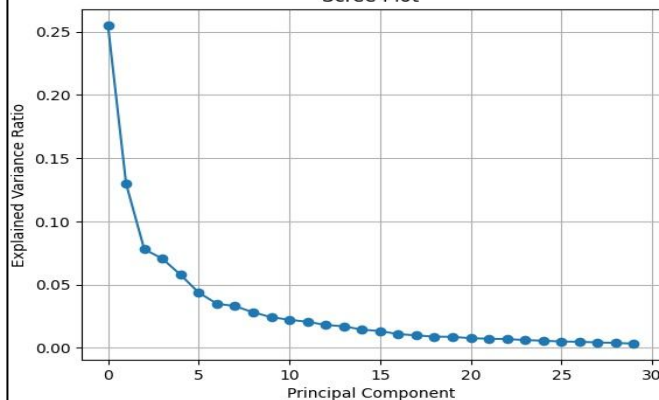
2. Right Frontal

- a. Bipolar 8: F8-T4
- b. Bipolar 11: F4-C4

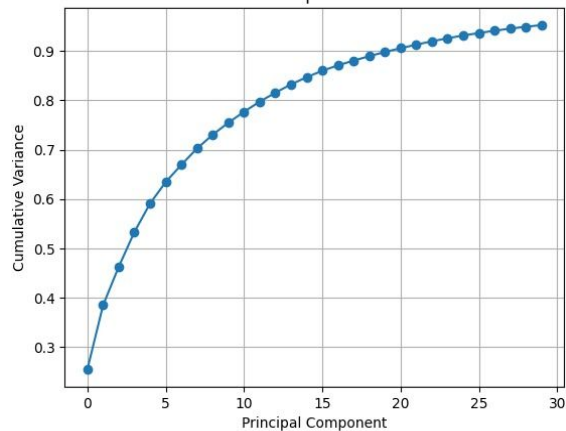
PCA:

- a. Original features: 132
- b. Features after PCA: 30

Scree Plot

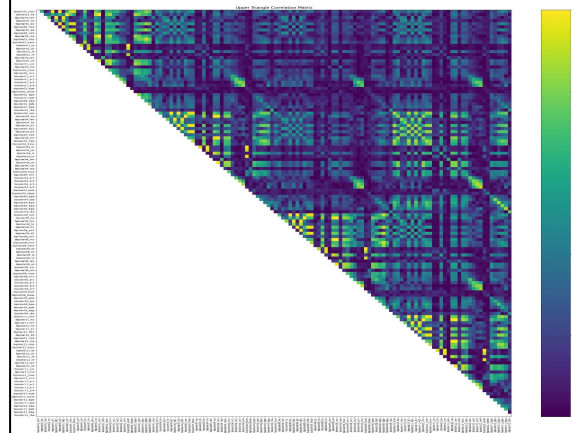


Cumulative Explained Variance



Correlation and Variance:

- a. Before: 132 features
- b. After: 77 features



Model Selection and Metrics

1. Models:

- Support Vector Machine (Linear Kernel)
- Support Vector Machine (RBF Kernel)
- Random Forest Classifier
- XGBoost Classifier
- K-Nearest Neighbors (KNN)

2. Evaluation Metrics:

- **Confusion Matrix:** Shows how many predictions were correct and incorrect for each class.
- **Accuracy:** Percentage of total predictions that were correct.
- **Precision:** Of all predicted positives, how many were actually positive.
- **Recall (Sensitivity):** Of all actual positives, how many were correctly detected.
- **F1 Score:** A balance between precision and recall.
- **AUC-ROC:** Measures how well the model separates the classes across different thresholds.
- **Balanced Accuracy:** Average of recall for each class, useful when classes are imbalanced.

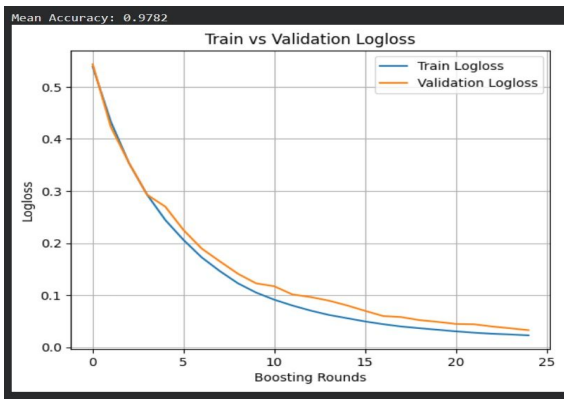
Results: Whole Brain Analysis

- The initial XGBoost model provided feature importance scores, allowing us to identify and keep only the most meaningful features.
 - Using a 95% cumulative importance threshold, the feature set was reduced to a smaller, more informative subset.
 - Retraining the model on this reduced set showed that the classifier maintained strong accuracy, confirming that the selected features carried the essential information.
-
- Running 50-fold stratified cross-validation produced 50 accuracy values, giving a thorough estimate of model stability.
 - The mean accuracy across all folds reflects how consistently the model performs on different data splits.
 - Logloss curves for training and validation help verify that the model learns steadily and does not overfit.

(352, 253) (152, 253) (352,) (152,)
Accuracy: 0.9802631578947368

Classification Report:				
	precision	recall	f1-score	support
0	0.99	0.97	0.98	71
1	0.98	0.99	0.98	81
accuracy			0.98	152
macro avg	0.98	0.98	0.98	152
weighted avg	0.98	0.98	0.98	152

Selected 49 features covering 95% importance.
Reduced dataset saved as 'EEG_XGBoost_selected_features.xlsx'
Accuracy on reduced features: 0.9603960396039604



Results : Region Wise Analysis - Correlation

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Frontal	SVM (Linear)	0.7632	0.7636	0.7647	0.763	0.7647	0.8321
	SVM (RBF)	0.8026	0.8023	0.8035	0.8023	0.8035	0.9138
	Random Forest	0.9539	0.9551	0.9568	0.9539	0.9568	0.9972
	XGBoost	0.9211	0.9216	0.9233	0.921	0.9233	0.9781
	KNN	0.9605	0.9602	0.9621	0.9605	0.9621	0.9901
Right Frontal	SVM (Linear)	0.6711	0.6723	0.6644	0.664	0.6644	0.7427
	SVM (RBF)	0.8026	0.8023	0.8035	0.8023	0.8035	0.8906
	Random Forest	0.9079	0.9075	0.9092	0.9078	0.9092	0.9591
	XGBoost	0.9079	0.9075	0.9075	0.9075	0.9075	0.948
	KNN	0.8684	0.8681	0.8696	0.8682	0.8696	0.9553
Frontal Region	SVM (Linear)	0.7368	0.7373	0.7383	0.7367	0.7383	0.8341
	SVM (RBF)	0.8224	0.8224	0.8238	0.8222	0.8238	0.912
	Random Forest	0.9539	0.9534	0.9551	0.9538	0.9551	0.996
	XGBoost	0.9013	0.9042	0.9048	0.9013	0.9048	0.9696
	KNN	0.9408	0.9408	0.9427	0.9407	0.9427	0.9868

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Temporal	SVM (Linear)	0.7829	0.7822	0.7833	0.7824	0.7833	0.8532
	SVM (RBF)	0.8421	0.8466	0.8379	0.8399	0.8379	0.9178
	Random Forest	0.875	0.8772	0.8723	0.8738	0.8723	0.9469
	XGBoost	0.8684	0.8714	0.8652	0.8669	0.8652	0.9428
	KNN	0.8618	0.8683	0.8573	0.8596	0.8573	0.9364
Right Temporal	SVM (Linear)	0.8355	0.8347	0.8352	0.8349	0.8352	0.8957
	SVM (RBF)	0.8947	0.8952	0.8969	0.8947	0.8969	0.9456
	Random Forest	0.8816	0.8809	0.8819	0.8812	0.8819	0.9565
	XGBoost	0.9079	0.9075	0.9075	0.9075	0.9075	0.9663
	KNN	0.9013	0.9012	0.9005	0.9008	0.9005	0.9618
Temporal Region	SVM (Linear)	0.8553	0.8549	0.8564	0.855	0.8564	0.9238
	SVM (RBF)	0.9079	0.9072	0.9084	0.9076	0.9084	0.9567
	Random Forest	0.9474	0.9471	0.9471	0.9471	0.9471	0.9857
	XGBoost	0.9408	0.9409	0.9401	0.9405	0.9401	0.9833
	KNN	0.9342	0.9386	0.9313	0.9335	0.9313	0.989

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Parietal	SVM (Linear)	0.7763	0.7808	0.771	0.7724	0.771	0.9016
	SVM (RBF)	0.9079	0.9144	0.904	0.9066	0.904	0.9602
	Random Forest	0.9211	0.9279	0.9172	0.9199	0.9172	0.9717
	XGBoost	0.9211	0.9252	0.9181	0.9202	0.9181	0.9753
	KNN	0.9539	0.9542	0.9533	0.9537	0.9533	0.9917
Right Parietal	SVM (Linear)	0.75	0.7505	0.7515	0.7498	0.7515	0.8473
	SVM (RBF)	0.8553	0.8629	0.8503	0.8527	0.8503	0.9503
	Random Forest	0.8421	0.8446	0.8388	0.8403	0.8388	0.9247
	XGBoost	0.8421	0.8446	0.8388	0.8403	0.8388	0.9195
	KNN	0.9145	0.9156	0.9128	0.9138	0.9128	0.9718
Parietal Region	SVM (Linear)	0.8421	0.8446	0.8388	0.8403	0.8388	0.9277
	SVM (RBF)	0.9342	0.9415	0.9304	0.9333	0.9304	0.9772
	Random Forest	0.9211	0.9252	0.9181	0.9202	0.9181	0.9877
	XGBoost	0.9145	0.9197	0.9111	0.9134	0.9111	0.9908
	KNN	0.9605	0.9599	0.9612	0.9604	0.9612	0.9943

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Occipital	SVM (Linear)	0.8289	0.8287	0.8273	0.8279	0.8273	0.9037
	SVM (RBF)	0.9342	0.9339	0.9339	0.9339	0.9339	0.9746
	Random Forest	0.9474	0.9471	0.9471	0.9471	0.9471	0.9851
	XGBoost	0.9342	0.9339	0.9339	0.9339	0.9339	0.9769
	KNN	0.9408	0.9409	0.9401	0.9405	0.9401	0.9837
Right Occipital	SVM (Linear)	0.8224	0.8333	0.8159	0.8182	0.8159	0.9077
	SVM (RBF)	0.9408	0.9422	0.9392	0.9404	0.9392	0.9744
	Random Forest	0.9013	0.9023	0.8996	0.9006	0.8996	0.9678
	XGBoost	0.9013	0.9007	0.9022	0.9011	0.9022	0.969
	KNN	0.8882	0.888	0.8872	0.8876	0.8872	0.9744
Occipital Region	SVM (Linear)	0.8684	0.8685	0.867	0.8676	0.867	0.9428
	SVM (RBF)	0.9408	0.9403	0.941	0.9406	0.941	0.9783
	Random Forest	0.9474	0.9468	0.948	0.9472	0.948	0.9761
	XGBoost	0.9276	0.9289	0.926	0.9271	0.926	0.9748
	KNN	0.9276	0.9276	0.9295	0.9276	0.9295	0.983

Results : Region Wise Analysis - PCA

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Frontal	SVM Linear	0.7303	0.7296	0.7304	0.7297	0.7304	0.8305
	SVM RBF	0.8289	0.8286	0.8299	0.8287	0.8299	0.9369
	Random Forest	0.8158	0.8163	0.8176	0.8157	0.8176	0.9142
	XGBoost	0.7763	0.7768	0.778	0.7762	0.778	0.8837
	KNN	0.8816	0.8835	0.8845	0.8816	0.8845	0.9637
Right Frontal	SVM Linear	0.6184	0.6161	0.6142	0.6141	0.6142	0.7036
	SVM RBF	0.875	0.8743	0.8749	0.8746	0.8749	0.9329
	Random Forest	0.7566	0.7556	0.7551	0.7553	0.7551	0.8593
	XGBoost	0.7697	0.7693	0.7674	0.768	0.7674	0.8484
	KNN	0.7763	0.778	0.7788	0.7763	0.7788	0.9012
Frontal Region	SVM Linear	0.6842	0.6838	0.6846	0.6837	0.6846	0.7981
	SVM RBF	0.9145	0.9145	0.9137	0.914	0.9137	0.9717
	Random Forest	0.8092	0.8092	0.8106	0.809	0.8106	0.9173
	XGBoost	0.8026	0.8023	0.8035	0.8023	0.8035	0.8985
	KNN	0.8618	0.8646	0.8652	0.8618	0.8652	0.9607

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Temporal	SVM Linear	0.8092	0.8092	0.8071	0.8078	0.8071	0.879
	SVM RBF	0.8487	0.8479	0.8485	0.8482	0.8485	0.9092
	Random Forest	0.8355	0.8388	0.8318	0.8334	0.8318	0.9257
	XGBoost	0.8421	0.8446	0.8388	0.8403	0.8388	0.932
	KNN	0.8355	0.8388	0.8318	0.8334	0.8318	0.9006
Right Temporal	SVM Linear	0.8355	0.8382	0.8387	0.8355	0.8387	0.8856
	SVM RBF	0.8816	0.8809	0.8819	0.8812	0.8819	0.9438
	Random Forest	0.8553	0.8546	0.8546	0.8546	0.8546	0.9048
	XGBoost	0.8421	0.8417	0.8432	0.8419	0.8432	0.9162
	KNN	0.8421	0.8417	0.8432	0.8419	0.8432	0.9355
Temporal Region	SVM Linear	0.8553	0.8558	0.8572	0.8552	0.8572	0.9033
	SVM RBF	0.9211	0.9207	0.9207	0.9207	0.9207	0.9774
	Random Forest	0.8816	0.8873	0.8776	0.8799	0.8776	0.9377
	XGBoost	0.875	0.8772	0.8723	0.8738	0.8723	0.9381
	KNN	0.9079	0.9083	0.9066	0.9073	0.9066	0.9776

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Parietal	SVM Linear	0.7961	0.7951	0.7956	0.7953	0.7956	0.8885
	SVM RBF	0.8947	0.9008	0.8908	0.8932	0.8908	0.9663
	Random Forest	0.8421	0.8414	0.8414	0.8414	0.8414	0.9184
	XGBoost	0.8553	0.8552	0.8538	0.8544	0.8538	0.9278
	KNN	0.8947	0.8944	0.896	0.8946	0.896	0.9703
Right Parietal	SVM Linear	0.7303	0.7326	0.733	0.7303	0.733	0.8194
	SVM RBF	0.8882	0.889	0.8864	0.8873	0.8864	0.9586
	Random Forest	0.8224	0.8301	0.8168	0.8189	0.8168	0.8859
	XGBoost	0.7829	0.7835	0.7798	0.7808	0.7798	0.8713
	KNN	0.9013	0.9023	0.8996	0.9006	0.8996	0.9604
Parietal Region	SVM Linear	0.8289	0.8294	0.8308	0.8288	0.8308	0.9184
	SVM RBF	0.9211	0.9252	0.9181	0.9202	0.9181	0.9824
	Random Forest	0.8882	0.8928	0.8846	0.8867	0.8846	0.9726
	XGBoost	0.8947	0.8983	0.8917	0.8936	0.8917	0.9732
	KNN	0.9539	0.9534	0.9551	0.9538	0.9551	0.9854

Region	Model	Accuracy	Precision	Recall	F1 Score	Balanced Acc.	AUC-ROC
Left Occipital	SVM Linear	0.8158	0.815	0.8159	0.8153	0.8159	0.8894
	SVM RBF	0.9079	0.9098	0.911	0.9079	0.911	0.9769
	Random Forest	0.8947	0.8944	0.896	0.8946	0.896	0.9583
	XGBoost	0.8947	0.8967	0.8978	0.8947	0.8978	0.9661
	KNN	0.9276	0.9271	0.9278	0.9274	0.9278	0.985
Right Occipital	SVM Linear	0.8289	0.8311	0.8256	0.827	0.8256	0.8924
	SVM RBF	0.9276	0.9271	0.9278	0.9274	0.9278	0.9802
	Random Forest	0.8487	0.8491	0.8467	0.8476	0.8467	0.9353
	XGBoost	0.8684	0.8685	0.867	0.8676	0.867	0.9405
	KNN	0.8947	0.8951	0.8934	0.8941	0.8934	0.9516
Occipital Region	SVM Linear	0.8355	0.8347	0.8352	0.8349	0.8352	0.9048
	SVM RBF	0.9342	0.9339	0.9357	0.9341	0.9357	0.987
	Random Forest	0.8553	0.8549	0.8564	0.855	0.8564	0.949
	XGBoost	0.8816	0.8812	0.8828	0.8814	0.8828	0.9588
	KNN	0.9013	0.9042	0.9048	0.9013	0.9048	0.9761

Summary of Results

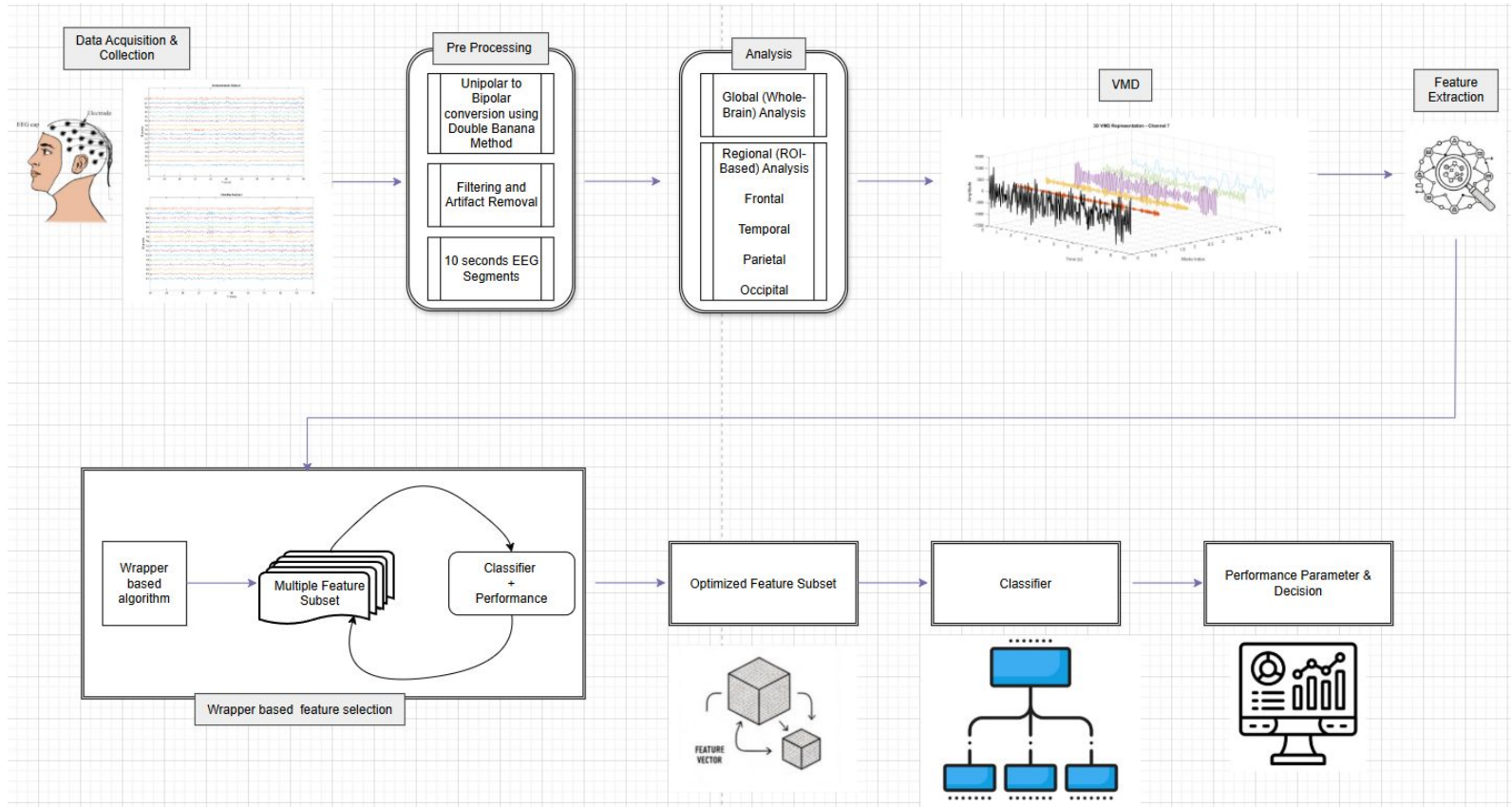
Correlation

Model	Region of Highest Accuracy(%)
SVM (Linear)	Occipital Region (86.84)
SVM (RBF)	Occipital Region (94.08)
Random Forest	Left Frontal (95.39)
XGBoost	Temporal Region (94.08)
KNN	Left Frontal / Parietal Region (96.05)

PCA

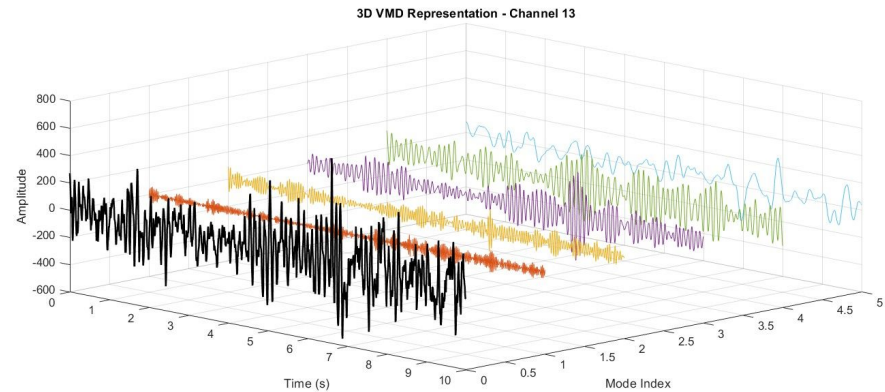
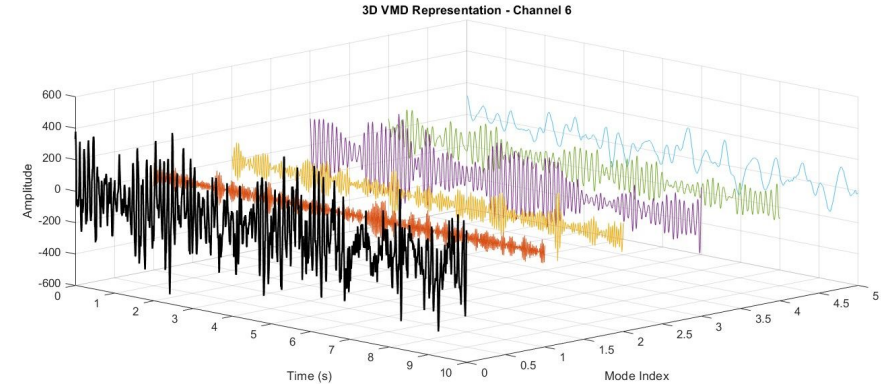
Model	Region of Highest Accuracy(%)
SVM (Linear)	Temporal Region (92.11)
SVM (RBF)	Occipital Region (93.42)
Random Forest	Left Occipital (89.47)
XGBoost	Parietal Region (89.47)
KNN	Parietal Region (95.34)

Method B



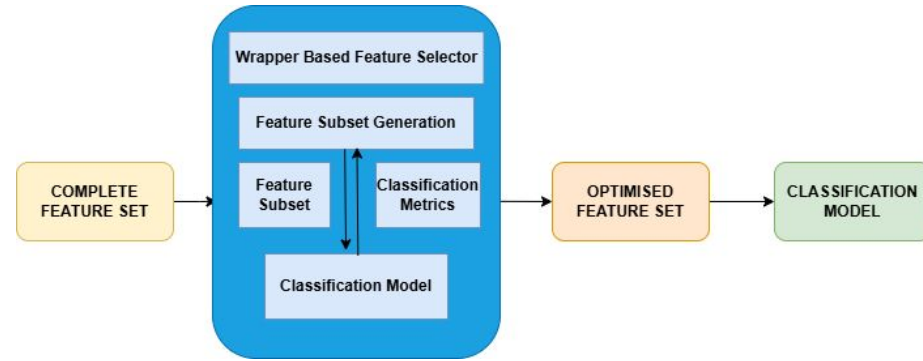
Variational Mode Decomposition

- VMD is a technique to handle the stable nature of the signal by decomposing the signals into its IMFs.
- VMD is reliable and robust, and it has been optimized for use with non-linear and non-stationary signals.
- VMD is widely used in fault diagnosis, biomedical signals, and financial data for precise signal decomposition.
- Designed for non-linear and non-stationary signals.
- Avoids mode mixing and spurious IMFs.
- Enables accurate time-frequency analysis.



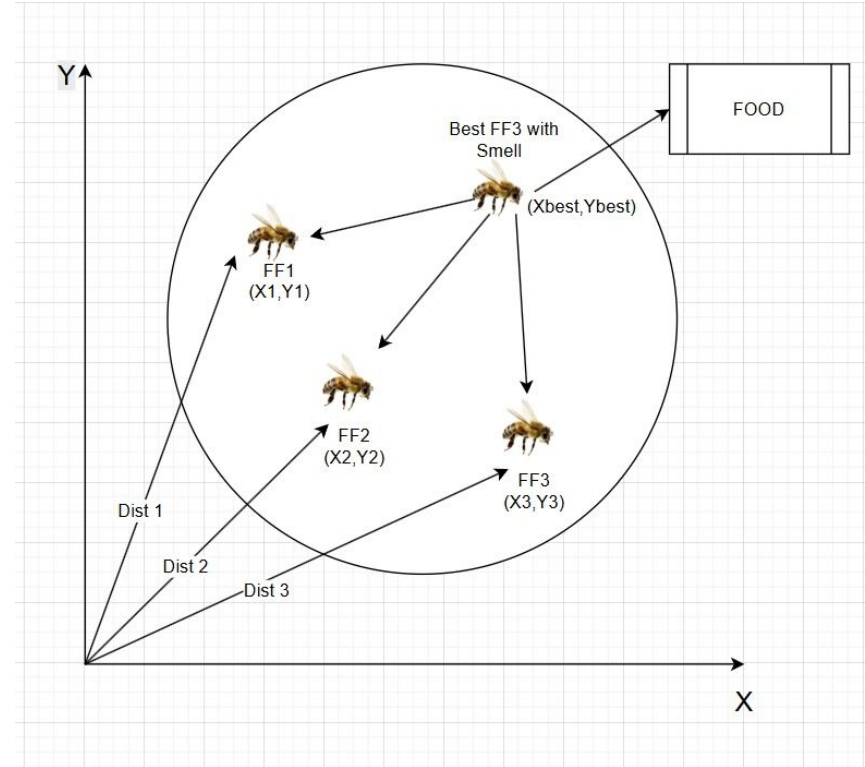
Wrapper Based Feature Selection

- After VMD, feature extraction is performed to obtain the complete feature vector.
- A wrapper-based feature selection method is then applied.
- This method uses algorithms to select an optimal subset of features.
- The selected feature subset is given to a classification model (KNN in our case).
- The model's performance is evaluated and fed back to the selection algorithm.
- Based on the results, a new feature subset is generated.
- This process continues in a loop until the termination criteria is satisfied.



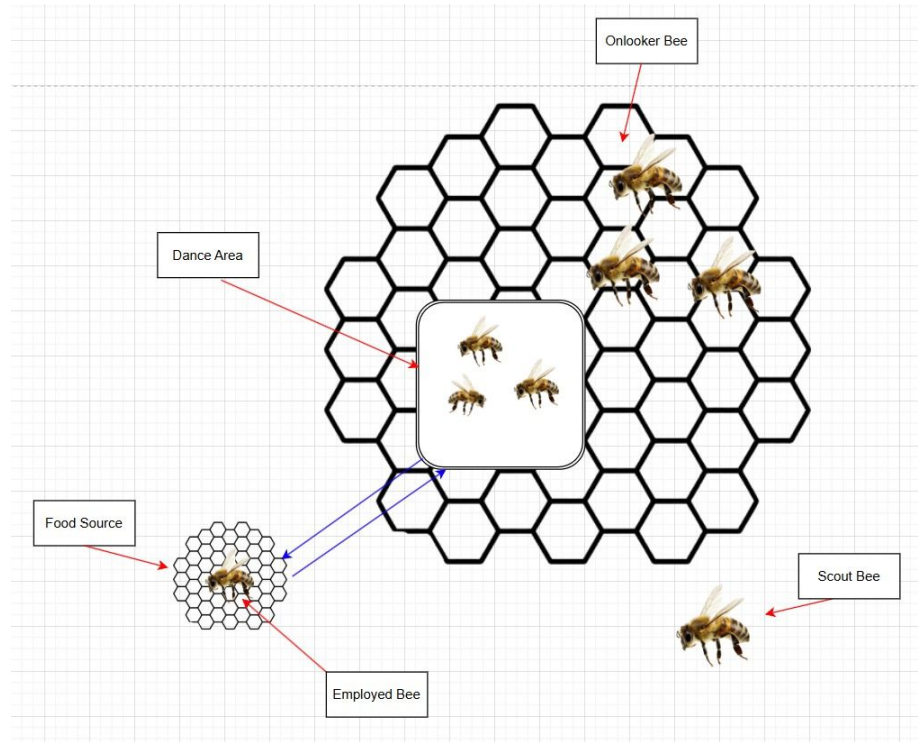
Optimisation Techniques - Fruit Fly Optimisation

- FOA is an optimization algorithm inspired by the food-searching behavior of fruit flies
- It uses smell and vision mechanisms to search for the best solution
- Solutions are initialized randomly and evaluated using a fitness function
- The algorithm iteratively moves towards the best solution found
- It continues updating solutions until an optimal or near-optimal result is achieved
- FOA is simple, fast, and easy to implement.



Optimisation Techniques - ABC Optimisation

- ABC is an optimization algorithm based on the intelligent behavior of honey bee swarms
- It consists of employed bees, onlooker bees, and scout bees working together
- Each solution is represented as a food source and its quality is measured as fitness
- Employed bees explore existing solutions while onlooker bees select better ones
- Scout bees search for new solutions to avoid local minima
- The algorithm balances exploration and exploitation to find optimal solutions
- It is effective for solving complex optimization problems



Results: Whole Brain Analysis

1. Optimisation algorithm: FOA

Model	Accuracy_mean	Accuracy_std	Precision_mean	Recall_mean	F1_mean	Balanced_Accuracy_mean	AUC_ROC_mean
RandomForest	0.9781584158	0.01458420883	0.9785539328	0.9776064686	0.9780211802	0.9776064686	0.9967519786
KNN	0.9662772277	0.01607301251	0.9673253327	0.9653458732	0.9660384757	0.9653458732	0.9909240415
SVM_rbf	0.9603168317	0.01084714808	0.9603656545	0.9600661253	0.9600983687	0.9600661253	0.9935759071
XGBoost	0.9405148515	0.02864152282	0.9412567212	0.9396169528	0.9401719731	0.9396169528	0.9874430397
SVM_linear	0.9146534653	0.02404452948	0.9144325579	0.9148987563	0.9142496107	0.9148987563	0.9735995477

2. Optimisation algorithm: ABC

Model	Accuracy_mean	Accuracy_std	Precision_mean	Recall_mean	F1_mean	Balanced_Accuracy_mean	AUC_ROC_mean
KNN	0.9742178218	0.01724951393	0.9742746749	0.9744543804	0.9741123704	0.9744543804	0.9939168123
SVM_rbf	0.9741980198	0.01012103984	0.9746237614	0.9742248261	0.9740570333	0.9742248261	0.9953969233
RandomForest	0.9642772277	0.01838135811	0.9651109712	0.9643214445	0.9640711654	0.9643214445	0.9955493884
XGBoost	0.9484554455	0.02286623749	0.950065125	0.9473464214	0.9480677161	0.9473464214	0.9895724124
SVM_linear	0.9404356436	0.01422791225	0.9425605014	0.9389265769	0.9398675036	0.9389265769	0.9802326378

Results: Region wise Analysis - FOA Algorithm

1	Region	Model	Accuracy_1	Precision_	Recall_me	F1_mean	Balanced_A	AUC_ROC_mean
2	Frontal	SVM_linear	0.782	0.785	0.784	0.781	0.784	0.858
3		SVM_rbf	0.837	0.840	0.838	0.836	0.838	0.919
4		RandomForest	0.895	0.896	0.893	0.894	0.893	0.951
5		XGBoost	0.857	0.859	0.855	0.856	0.855	0.933
6		KNN	0.859	0.861	0.862	0.859	0.862	0.935
7	Left Frontal	SVM_linear	0.687	0.693	0.690	0.686	0.690	0.765
8		SVM_rbf	0.768	0.770	0.769	0.767	0.769	0.859
9		RandomForest	0.800	0.801	0.801	0.799	0.801	0.884
10		XGBoost	0.786	0.786	0.785	0.785	0.785	0.870
11		KNN	0.794	0.797	0.797	0.793	0.797	0.868
12	Right frontal	SVM_linear	0.669	0.669	0.664	0.663	0.664	0.753
13		SVM_rbf	0.746	0.747	0.741	0.742	0.741	0.831
14		RandomForest	0.816	0.817	0.813	0.814	0.813	0.885
15		XGBoost	0.796	0.799	0.794	0.794	0.794	0.886
16		KNN	0.778	0.779	0.777	0.777	0.777	0.848

17	Region	Model	Accuracy_1	Precision_	Recall_me	F1_mean	Balanced_A	AUC_ROC_mean
18	Occipital	SVM_linear	0.865	0.866	0.864	0.864	0.864	0.916
19		SVM_rbf	0.895	0.896	0.895	0.894	0.895	0.949
20		RandomForest	0.861	0.861	0.861	0.860	0.861	0.940
21		XGBoost	0.875	0.877	0.874	0.874	0.874	0.944
22		KNN	0.861	0.863	0.864	0.861	0.864	0.936
23	Left Occipital	SVM_linear	0.798	0.799	0.798	0.797	0.798	0.876
24		SVM_rbf	0.855	0.860	0.854	0.854	0.854	0.918
25		RandomForest	0.863	0.864	0.864	0.863	0.864	0.938
26		XGBoost	0.845	0.847	0.844	0.844	0.844	0.924
27		KNN	0.833	0.839	0.833	0.832	0.833	0.925
28	Right Occipital	SVM_linear	0.800	0.803	0.796	0.797	0.796	0.885
29		SVM_rbf	0.833	0.837	0.832	0.832	0.832	0.914
30		RandomForest	0.845	0.847	0.844	0.844	0.844	0.916
31		XGBoost	0.851	0.854	0.851	0.850	0.851	0.922
32		KNN	0.814	0.815	0.814	0.813	0.814	0.888

33	Region	Model	Accuracy_1	Precision_	Recall_me	F1_mean	Balanced_A	AUC_ROC_mean
34	Parietal	SVM_linear	0.806	0.806	0.804	0.804	0.804	0.898
35		SVM_rbf	0.859	0.860	0.858	0.858	0.858	0.926
36		RandomForest	0.867	0.871	0.866	0.866	0.866	0.941
37		XGBoost	0.843	0.845	0.841	0.842	0.841	0.930
38		KNN	0.837	0.840	0.835	0.836	0.835	0.920
39	Left Parietal	SVM_linear	0.762	0.763	0.757	0.758	0.757	0.858
40		SVM_rbf	0.810	0.813	0.806	0.807	0.806	0.893
41		RandomForest	0.823	0.824	0.824	0.823	0.824	0.906
42		XGBoost	0.812	0.813	0.809	0.810	0.809	0.892
43		KNN	0.812	0.813	0.809	0.810	0.809	0.905
44	Right Parietal	SVM_linear	0.748	0.747	0.746	0.746	0.746	0.829
45		SVM_rbf	0.817	0.817	0.818	0.817	0.818	0.891
46		RandomForest	0.835	0.835	0.836	0.835	0.836	0.922
47		XGBoost	0.843	0.845	0.841	0.842	0.841	0.922
48		KNN	0.790	0.793	0.785	0.786	0.785	0.868

49	Region	Model	Accuracy_1	Precision_	Recall_me	F1_mean	Balanced_A	AUC_ROC_mean
50	Temporal	SVM_linear	0.831	0.835	0.832	0.830	0.832	0.911
51		SVM_rbf	0.857	0.859	0.857	0.856	0.857	0.937
52		RandomForest	0.847	0.848	0.848	0.847	0.848	0.936
53		XGBoost	0.851	0.853	0.849	0.850	0.849	0.928
54		KNN	0.877	0.878	0.877	0.876	0.877	0.944
55	Left Temporal	SVM_linear	0.679	0.679	0.678	0.677	0.678	0.769
56		SVM_rbf	0.724	0.727	0.724	0.722	0.724	0.810
57		RandomForest	0.752	0.755	0.750	0.750	0.750	0.840
58		XGBoost	0.788	0.790	0.787	0.787	0.787	0.861
59		KNN	0.740	0.745	0.738	0.737	0.738	0.815
60	RighthTemporal	SVM_linear	0.760	0.763	0.763	0.760	0.763	0.843
61		SVM_rbf	0.782	0.786	0.785	0.781	0.785	0.854
62		RandomForest	0.796	0.797	0.797	0.795	0.797	0.872
63		XGBoost	0.786	0.786	0.785	0.785	0.785	0.882
64		KNN	0.762	0.764	0.764	0.762	0.764	0.833

Results: Region wise Analysis - ABC Algorithm

Region	Model	Accuracy_f	Precision_n	Recall_rr	F1_mean	Balanced_A	AUC_ROC_mean
Frontal	SVM_linear	0.770	0.771	0.770	0.769	0.770	0.860
	SVM_rbf	0.851	0.851	0.851	0.850	0.851	0.929
	RandomForest	0.883	0.884	0.882	0.882	0.882	0.951
	XGBoost	0.859	0.862	0.857	0.858	0.857	0.938
	KNN	0.865	0.868	0.868	0.865	0.868	0.944
Left Frontal	SVM_linear	0.760	0.761	0.762	0.760	0.762	0.838
	SVM_rbf	0.794	0.795	0.795	0.793	0.795	0.878
	RandomForest	0.798	0.797	0.797	0.797	0.797	0.891
	XGBoost	0.821	0.824	0.818	0.819	0.818	0.897
	KNN	0.807	0.811	0.811	0.807	0.811	0.889
Right frontal	SVM_linear	0.700	0.703	0.695	0.695	0.695	0.758
	SVM_rbf	0.796	0.799	0.792	0.793	0.792	0.861
	RandomForest	0.843	0.846	0.841	0.841	0.841	0.909
	XGBoost	0.827	0.830	0.825	0.826	0.825	0.899
	KNN	0.804	0.808	0.803	0.802	0.803	0.868

Region	Model	Accuracy_f	Precision_n	Recall_rr	F1_mean	Balanced_A	AUC_ROC_mean
Parietal	SVM_linear	0.829	0.832	0.827	0.828	0.827	0.896
	SVM_rbf	0.873	0.873	0.872	0.872	0.872	0.937
	RandomForest	0.877	0.878	0.876	0.876	0.876	0.951
	XGBoost	0.865	0.869	0.864	0.864	0.864	0.947
	KNN	0.851	0.853	0.849	0.850	0.849	0.927
Left Parietal	SVM_linear	0.803	0.804	0.801	0.801	0.801	0.889
	SVM_rbf	0.837	0.840	0.834	0.835	0.834	0.917
	RandomForest	0.845	0.846	0.843	0.844	0.843	0.926
	XGBoost	0.839	0.843	0.836	0.837	0.836	0.928
	KNN	0.863	0.865	0.861	0.862	0.861	0.938
Right Parietal	SVM_linear	0.786	0.785	0.785	0.784	0.785	0.871
	SVM_rbf	0.817	0.817	0.816	0.816	0.816	0.896
	RandomForest	0.835	0.837	0.835	0.835	0.835	0.929
	XGBoost	0.835	0.836	0.835	0.834	0.835	0.921
	KNN	0.829	0.833	0.825	0.827	0.825	0.904

Region	Model	Accuracy_f	Precision_n	Recall_rr	F1_mean	Balanced_A	AUC_ROC_mean
Occipital	SVM_linear	0.861	0.865	0.861	0.860	0.861	0.920
	SVM_rbf	0.883	0.885	0.883	0.882	0.883	0.952
	RandomForest	0.871	0.872	0.872	0.871	0.872	0.948
	XGBoost	0.877	0.879	0.877	0.876	0.877	0.945
	KNN	0.869	0.871	0.871	0.869	0.871	0.934
Left Occipital	SVM_linear	0.788	0.790	0.787	0.787	0.787	0.882
	SVM_rbf	0.825	0.827	0.827	0.825	0.827	0.919
	RandomForest	0.861	0.864	0.862	0.861	0.862	0.938
	XGBoost	0.861	0.861	0.861	0.860	0.861	0.932
	KNN	0.841	0.841	0.842	0.841	0.842	0.910
Right Occipital	SVM_linear	0.806	0.806	0.805	0.804	0.805	0.892
	SVM_rbf	0.861	0.864	0.861	0.860	0.861	0.921
	RandomForest	0.825	0.828	0.824	0.824	0.824	0.915
	XGBoost	0.847	0.850	0.845	0.846	0.845	0.925
	KNN	0.814	0.814	0.814	0.813	0.814	0.883

Region	Model	Accuracy_f	Precision_n	Recall_rr	F1_mean	Balanced_A	AUC_ROC_mean
Temporal	SVM_linear	0.831	0.835	0.832	0.830	0.832	0.911
	SVM_rbf	0.857	0.859	0.857	0.856	0.857	0.937
	RandomForest	0.847	0.848	0.848	0.847	0.848	0.936
	XGBoost	0.851	0.853	0.849	0.850	0.849	0.928
	KNN	0.877	0.878	0.877	0.876	0.877	0.944
Left Temporal	SVM_linear	0.690	0.694	0.691	0.688	0.691	0.785
	SVM_rbf	0.766	0.767	0.765	0.764	0.765	0.839
	RandomForest	0.796	0.797	0.795	0.795	0.795	0.869
	XGBoost	0.780	0.782	0.778	0.778	0.778	0.867
	KNN	0.760	0.761	0.760	0.759	0.760	0.830
RighTemporal	SVM_linear	0.810	0.811	0.811	0.809	0.811	0.886
	SVM_rbf	0.835	0.837	0.838	0.835	0.838	0.914
	RandomForest	0.827	0.829	0.828	0.827	0.828	0.901
	XGBoost	0.828	0.828	0.827	0.827	0.827	0.910
	KNN	0.849	0.850	0.851	0.849	0.851	0.918

Summary of Results

FOA Optimisation

Model	Region of Highest Accuracy (%)
SVM_linear	Occipital (86.50%)
SVM_rbf	Occipital (89.49%)
RandomForest	Frontal (89.50%)
XGBoost	Occipital (87.50%)
KNN	Temporal (87.70%)

ABC Optimisation

Model	Region of Highest Accuracy (%)
SVM_linear	Occipital (86.10%)
SVM_rbf	Occipital (88.29%)
RandomForest	Frontal (88.30%)
XGBoost	Occipital (87.70%)
KNN	Temporal (87.70%)

SDG Goals

SDG 3: Good Health and Well-being :

- This project aligns with SDG 3: Good Health and Well-being by supporting early and objective detection of schizophrenia through EEG-based biomarkers. It promotes faster diagnosis, better clinical decisions, and improved access to treatment using a non-invasive and affordable technology-driven assessment. The work contributes to strengthening mental health care and ensuring timely support for individuals affected by psychiatric disorders.



Fig 6. SDG 3 – Good Health and Well-Being

References

- [1] National Institute of Mental Health, “Schizophrenia,” [Online]. Available: <https://www.nimh.nih.gov/health/statistics/schizophrenia>. [Accessed: 28-Aug-2025].
- [2] G. Ayano et al., “Misdiagnosis, detection rate, and associated factors of severe psychiatric disorders in specialized psychiatry centers in Ethiopia,” [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7856725/>. [Accessed: 28-Aug-2025].
- [3] “An interpretable XAI deep EEG model for schizophrenia diagnosis using feature selection and attention mechanisms,” [Online]. Available: <https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2025.1630291/full>[Accessed: 28-Aug-2025].
- [4] “Practical AI application in psychiatry: historical review and future directions,” [Online]. Available: <https://www.nature.com/articles/s41380-025-03072-3>. [Accessed: 28-Aug-2025].
- [5] “High vs low frequency neural oscillations in schizophrenia,” [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3122299/>. [Accessed: 28-Aug-2025].
- [6] “EEG data of healthy adolescents and adolescents with symptoms of schizophrenia,” [Online]. Available: http://brain.bio.msu.ru/eeg_schizophrenia.htm. [Accessed: 28-Aug-2025].
- [7] “miRNA-Based Diagnosis of Schizophrenia Using Machine Learning,” [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11900116/>. [Accessed: 28-Aug-2025].

References

- [8] “Structural MRI-Based Schizophrenia Classification Using Complex Feature Extraction and Deep Learning,” [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9139344/>. [Accessed: 28-Aug-2025].
- [9] “Diagnosis of Schizophrenia Using Feature Extraction from EEG Time-Frequency Images,” [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12292799/>. [Accessed: 28-Aug-2025].
- [10] “Three-Dimensional Convolutional Autoencoder Extracts Features of Psychiatric Disorders,” [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8294943/>. [Accessed: 28-Aug-2025].
- [11] “Montages and Technical Components,” *LearningEEG*, [Online]. Available: <https://www.learningeeg.com/montages-and-technical-components>. [Accessed: 09-Oct-2025].
- [12] “EEG Feature Extraction Toolbox,” *GitHub*, [Online]. Available: <https://github.com/JingweiToo/EEG-Feature-Extraction-Toolbox>. [Accessed: 02-Dec-2025].
- [13] “Trends in EEG Machine Learning Research,” *KDD*, [Online]. Available: <https://kdd.org/kdd2023/wp-content/uploads/2023/08/murungi2023trends.pdf>. [Accessed: 02-Dec-2025].
- [14] “Lobes of the Brain,” AliLa Medical Media - Neurology, [Online]. Available: <https://www.alilamedicalmedia.com/-/galleries/images-videos-by-medical-specialties/neurology/-/medias/56de46a4-087f-11e3-bc56-63d4f0448c9c-lobes-of-the-brain> [Accessed: 02-Dec-2025].
- [15] “Sleep Stage Classification Using Variational Mode Decomposition and Wrapper-Based Feature Selection From the Single Channel EEG” [online]. Available : <https://ieeexplore.ieee.org/abstract/document/11071537> [Accessed: 02-Dec-2025]

References

- [16] “SDG 3 – Good Health and Well-Being,” SPIHER – Sri Pushpam Institute of Higher Education and Research, [Online]. Available: <https://spiher.ac.in/sdg-3-good-health-and-well-being/>. [Accessed: 02-Dec-2025].
- [17] “A New Fruit Fly Optimization Algorithm: Taking the Financial Distress Model as an Example,” IEEE, [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/9187840>. [Accessed: 02-Dec-2025].
- [18] “A Powerful and Efficient Algorithm for Numerical Function Optimization: Artificial Bee Colony (ABC) Algorithm,” Springer, [Online]. Available: <https://link.springer.com/article/10.1007/S10898-007-9149-X>. [Accessed: 02-Dec-2025].